

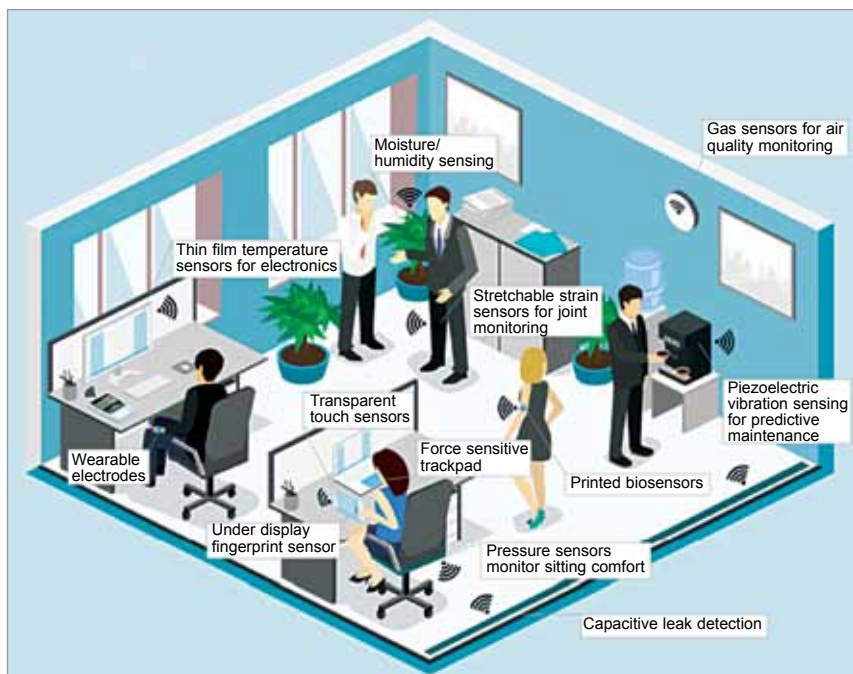


Fig. 2: Various sensing features in an office space (Credit: IDTechEx)

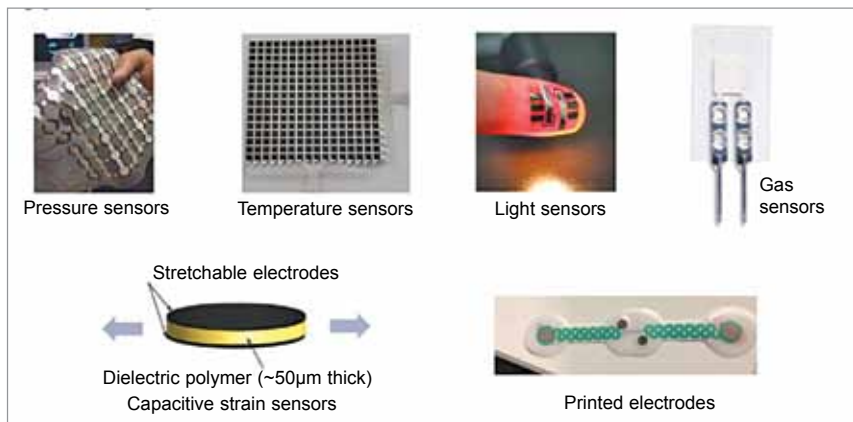


Fig. 3: Different types of printed flexible sensors (Credit: IDTechEx)

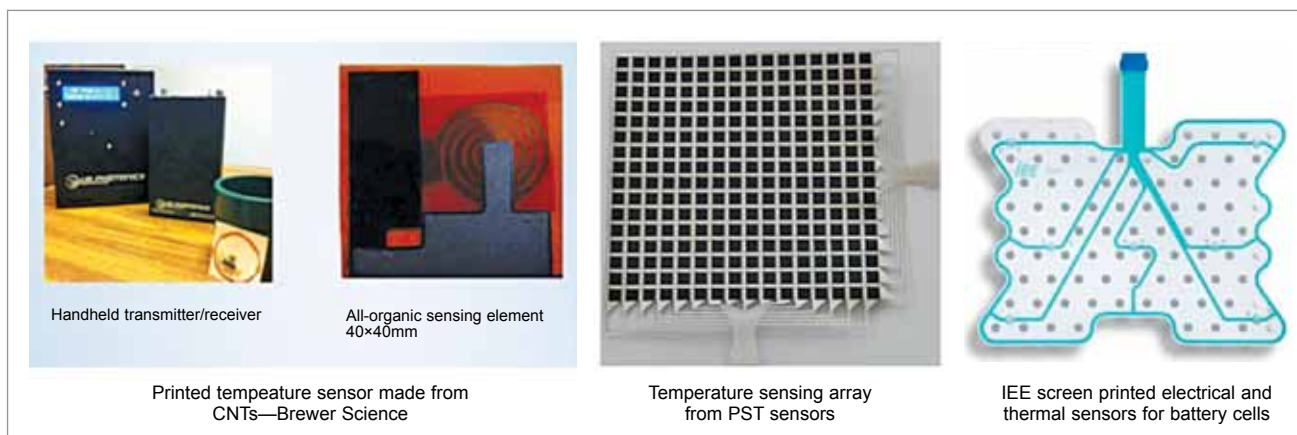


Fig. 4: Different types of spatial arrayed temperature sensors (Credit: IDTechEx)

enable monitoring of medical rehabilitation or can be incorporated in gloves for augmented reality. With many compelling applications that provoke curiosity, let's take a deeper look at some of the applications.

Spatially resolved temperature sensors. Thermistors and RTDs are generally quite small, inexpensive, and adept function-wise. Hence, printed electronics is preferred as it enables a spatial array of temperature sensors. While the amount of charge stored in a battery depends a lot on temperature, these sensors are well suited for implementing battery monitoring in mobile phones and electric vehicles. The range of batteries can be significantly increased by making appropriate adjustments in temperature. Subsequently, the sensor can also alert any problems occurring in the battery that can cause heat spikes.

Healthcare monitoring

There are applications that include monitoring compact electronics with the help of a thin spatially arrayed temperature sensor for more efficient tracking and packaging.

Healthcare monitoring is another place where you can wrap a flexible, comfortable temperature sensor around your arms or legs and monitor the healing process. This induces a change in temperature due to the increased blood flow.